

Climate Change Data Analysis

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December 8, 2025

The climate has changed and it is always changing.
- Raj Shah, White House Principal Deputy Press Secretary

1 Introduction

In this project, we attempt to attach a meaning to Mr. Shah's statement that climate always changes. By interpreting 'climate' as average global temperature and 'change' as the climate's statistical behavior, we can perform an analysis. For our analysis, we worked with two global temperature datasets, a Land-Ocean index and a Global Station index. The data spans from 1880 to 2025. Each month was treated as an independent temperature measurement over the 145-year span. The program reads 1740 monthly values, each a measure of deviation from a 1951–1980 temperature average. Our initial hypothesis was that climate does always change, but we were eager to quantify the idea. To determine if the climate always changes, we will analyze the monthly temperature deviations over time. If the deviations consistently change over time, we will conclude that climate exhibits directional change beyond natural variability.

To summarize our process, we first cleaned the datasets, turning the text files of monthly deviations into an array we could work with in a .cpp file. Next, we examined the time intervals between temperature records, storing gap years and gap sizes as variables in another array. Finally, we performed a linear regression analysis on each of the 12 months independently for both datasets. First, we analyzed the size of the gaps between temperature records over time. Second, we used the data to test for global warming, finding the warming rate over time. Our final program outputs a series of best fit line equations (see subsections 4.1 and 4.2). For every month in both data sets, we fit an equation to how gap sizes have changed, and how temperatures have deviated from the average, over the 145-year span.

As we will prove in section 3, the climate does always change, but the change is directional. While all months experience natural year-to-year variability, our data analysis highlights two overarching trends. The gaps between temperature records get smaller over time for 11/12 months in the Land-Ocean data and 12/12 months in the Global Station data. Meanwhile, all 12 months in both datasets exhibited an increased warming over time, visualized in the Appendix (section 7).

2 Reading the Data

We analyzed two of NASA's long-term global temperature datasets: `GLB.Ts.txt` (Global Station temperature indexes) and `GLB.Ts+dSST.txt` (Land-Ocean temperature indexes). Deviations from the 1951-1980 average are given for each month of each year in the range January 1880 – October 2025. Initially, both text files contained lines that were not necessary for our analysis, so we opted to use `grep` to clean the data and create two new files, `GLB.Ts_clean.txt` and `GLB.Ts+dSST_clean.txt`. We took out ancillary columns that were not necessary for our analysis, such as `J-D`, `D-N`, `DJF`, `MAM`, `JJA`, `SON`, and an extra `Year` column. Next, to convert the text into a variable C++ could interact with, we made a list of the available years, and an array `monthly_deviation` of size `12 x nyears`, where `nyears` is the number of full years in the file. To ensure the data analysis only uses full years, the program traverses the array and counts the elements in each year. If `count < 12`, the year is removed from the `nyears` total and the elements are removed from the larger `monthly_deviation` array.

Example Year - 1880 Land-Ocean Deviations
(first row of the Land-Ocean `monthly_deviation` array)

Jan	Fab	Mar	Apr	May	Jun	Jul	Aug	Sep	Oc	Nov	Dec
-19	-25	-9	-17	-10	-21	-19	-11	-15	-24	-23	-18

3 Evaluating the Statistical Hypothesis

Now that we've extracted the data, we can begin our data analysis. In our program, we aim to quantify whether climate always changes by analyzing the statistical behavior of global temperature data from 1880 to 2025. Our program consists of three main components: 1) creating an array of previous records, 2) identifying when temperature records were broken and calculating the gaps between consecutive records, and 3) fitting linear regression models to each calendar month independently to analyze warming trends, and whether gap sizes are increasing, decreasing, or remaining constant over time. From the code output, we can begin to visualize the climate behavior and distinguish between natural variability and a systematic, directional change.

3.1 Creating an Array of Previous Records

From Mr. Shah's claim, we assume that climate has a stationary distribution, so highs and lows have a probability distribution independent of time. For n data points, each point has a chance of $1/n$ to be a record high. Over $n + 1$ years each year has a chance of $1/(n + 1)$, and the $n + 1$ st year has a chance $1/(n + 1)$ of being a record. As a function of n , the chance of a record high decreases as $1/n$, and the gap between successive highs is approximately a linear function of the year.¹

To test this in our program, we made an array of `previous_record` that is the same shape as `monthly_deviation`. Treating months as independent measurements, this array records whether that year was a record or, if not, when the previous record occurred, allowing us to account for gaps between the records which we will analyze in section 3.2. To create this array, we developed a function, `prevRecord`, that iterates through the temperature deviation data chronologically and flags each month that exceeds all previous temperature measurements. This keeps an ongoing record of when global temperature records were broken, which is then used to calculate the time intervals, or gaps, between consecutive record-breaking years. The function header for `prevRecord` is shown below:

```
void prevRecord(vector<int> nyears, vector<int> dev, vector<int>& previousRecord)
```

3.2 Finding Gaps Between Records

Using the `previous_record` array, we computed the gaps between the record highs for each month. We did this through the use of a function called `gaps` which takes five parameters. It takes in the specific month we are working with, the `nyears` vector to indicate when measurements have occurred, the `previous_record` vector to indicate previous records, a `gapyears` vector, and a `gapsizes` vector. The `gapyears` vector stores the years when records occur and the `gapsizes` vector stores the number of years between consecutive records. For each month, the program traverses through every year in the dataset and records every year a record was set. Once a new record occurs, we computed the time interval between that record and the previous one. That time interval value is stored in `gapsizes`, and the record year is stored in `gapyears`. This data collection and storage informs the eventual output, a record of how often the new highs occurred each month. The function header for `gaps` is shown below:

```
void gaps(int month, vector<int> nyears, vector<int> previousRecord, vector<int>& gapyears, vector<int>& gapsizes);
```

¹The Art of HPC, Volume 3: Introduction to Scientific Programming, by Victor Eijkhout. (2017-2025).

3.3 Finding the Best Fit Line

Our program uses linear regression to visualize trends, the frequency of temperature records and the rate of warming, for each respective month. To find the new best line of fit, we used the following equations: ²

$$b = \frac{SS_{xy} - S_x S_y}{\Delta}$$
$$m = \frac{SS_{xy} - S_x S_y}{\Delta}$$
$$\Delta = SS_{xx} - (S_x)^2$$

We used the following elements:

- S: number of values
- Sx: sum of x-values
- Sy: sum of y-values
- Sxx: sum of the square of x-values
- Sxy: sum of the product of x-values and y-values

From these equations, we could develop the `linearFunc` function which performs linear regression on provided x and y data vectors, calculating the slope and y-intercept of the best-fit line. The second line calls `linearFunc` to perform linear regression on the gap data for a current month. This intends to create a best fit line out of the frequency of temperature records. The x-values (`xgap`) represent the records (1st record, 2nd record...), while the y-values (`ygap`) represent the gap sizes in years between consecutive records. The resulting slope (`mgap`) reveals whether gaps are increasing, decreasing, or staying constant over time.

```
void linearFunc(vector<int> x, vector<int> y, double& m, double& b)

linearFunc(xgap, ygap, mgap, bgap)
```

Initially, `xgap` and `ygap` contain the prepared data points, and `mgap` and `bgap` are initialized to 0.0. After the `linearFunc` call, `mgap` contains the rate of change in gap size per record and `bgap` contains the y-intercept. If `mgap` is > 0 , the gaps between records are increasing. If `mgap` is < 0 , the gaps between records are decreasing.

After doing the regression analysis, we came to the conclusion that the gaps are not linearly increasing. (see monthly regression in 4.1 and 4.2). This contradicts the stationary distribution that Mr. Shah was referring to. In a stationary climate, the gaps should grow over time. This would produce a positive slope, but our data produced a negative slope for nearly every month (`mgap` < 0 for 95.8% of data). This indicates record highs are happening at an increasing rate rather than decreasing. We were therefore driven to find a trend in deviations. To find a trend between deviation and time, we performed a second regression, letting the x-values be the years and y-values be the temperature deviation vector. Once we ran the `linearFunc` function with these new values, we were able to see a positive trend of consistent warming.

4 Output Analysis

When we run the program, it reads in 145 years and 1740 monthly deviations, meaning all of the years had full data sets except for our current year, 2025. The code first outputs the first regression analysis and the second deviation analysis for the Land-Ocean dataset (4.1). Next, the program prints the regression analysis and deviation analysis for the the Global Station data (4.2). We will dive into the output for each data set below.

²Numerical Recipes: The Art of Scientific Computing, Third Edition, by W.H. Press, S.A. Teukolsky, W.T. Vetterling, and B.P. Flannery. Version 3.04 (2011).

4.1 Land-Ocean Temperature Data Output

```
Month 0 regression: gap size = -1.2028 * recordIndex + 19.8182
Month 1 regression: gap size = -0.660714 * recordIndex + 14.8857
Month 2 regression: gap size = -0.531868 * recordIndex + 14.2747
Month 3 regression: gap size = -1.00588 * recordIndex + 17.55
Month 4 regression: gap size = -1.18824 * recordIndex + 19.1
Month 5 regression: gap size = -0.645113 * recordIndex + 13.9737
Month 6 regression: gap size = -1.85495 * recordIndex + 24.1978
Month 7 regression: gap size = -0.941176 * recordIndex + 16.9412
Month 8 regression: gap size = -0.558308 * recordIndex + 13.2484
Month 9 regression: gap size = -0.29902 * recordIndex + 11.1029
Month 10 regression: gap size = -0.519298 * recordIndex + 12.7193
Month 11 regression: gap size = 0.206294 * recordIndex + 10.5758
```

The regression of each month shows the relation between time and gap sizes. 11 out of 12 months have negative slopes (-1.85 to -0.30), so gap sizes decrease over time. Only Month 11, December, shows a slightly positive slope of 0.21 . This means that, for most months, record-breaking temperatures occur more frequently in recent decades.

```
Month 0 trend: deviation = 0.874598 * t + -56.247
Month 1 trend: deviation = 0.865749 * t + -54.7202
Month 2 trend: deviation = 0.895627 * t + -55.092
Month 3 trend: deviation = 0.827165 * t + -52.8179
Month 4 trend: deviation = 0.794477 * t + -51.561
Month 5 trend: deviation = 0.803704 * t + -53.4529
Month 6 trend: deviation = 0.75535 * t + -47.5162
Month 7 trend: deviation = 0.784628 * t + -49.7277
Month 8 trend: deviation = 0.780472 * t + -48.9043
Month 9 trend: deviation = 0.78068 * t + -46.3952
Month 10 trend: deviation = 0.802082 * t + -48.7292
Month 11 trend: deviation = 0.812888 * t + -52.0452
```

The deviation of each month is a relation between the years and the warming rates. All 12 months have a positive slope (0.75 to 0.89), which shows a trend of consistent warming trend for each month.

4.2 Global Station Temperature Data Output

```
Month 0 regression: gap size = -1.56593 * recordIndex + 22.0385
Month 1 regression: gap size = -0.18022 * recordIndex + 11.6374
Month 2 regression: gap size = -0.434066 * recordIndex + 13.5
Month 3 regression: gap size = -1.38462 * recordIndex + 20.6703
Month 4 regression: gap size = -1.64118 * recordIndex + 22.95
Month 5 regression: gap size = -3.59394 * recordIndex + 34.0667
Month 6 regression: gap size = -0.373684 * recordIndex + 11.0737
Month 7 regression: gap size = -3.71329 * recordIndex + 36.1364
Month 8 regression: gap size = -0.243083 * recordIndex + 9.13439
Month 9 regression: gap size = -0.813725 * recordIndex + 15.7941
Month 10 regression: gap size = -0.383117 * recordIndex + 11.0238
Month 11 regression: gap size = -1.76264 * recordIndex + 23.5055
```

The regression of each month shows the relation between time and gap sizes. All 12 months have negative slopes (-3.71 to -0.18), so gap sizes decrease over time. This means that, for all months in the Global Station dataset, record-breaking temperatures occur more frequently in recent decades.

```

Month 0 trend: deviation = 1.13575 * t + -73.3048
Month 1 trend: deviation = 1.16404 * t + -75.4316
Month 2 trend: deviation = 1.14558 * t + -72.1579
Month 3 trend: deviation = 1.03433 * t + -65.6025
Month 4 trend: deviation = 1.03985 * t + -68.938
Month 5 trend: deviation = 1.02043 * t + -67.6017
Month 6 trend: deviation = 0.953472 * t + -58.1741
Month 7 trend: deviation = 0.946044 * t + -56.1634
Month 8 trend: deviation = 0.937652 * t + -54.7316
Month 9 trend: deviation = 0.966553 * t + -55.9987
Month 10 trend: deviation = 1.00312 * t + -60.9213
Month 11 trend: deviation = 1.01899 * t + -63.3051

```

The deviation of each month is a relation between the years and the warming rates. All 12 months have a positive slope (0.94 to 1.16), which shows a trend of consistent warming trend for each month. This was consistent with the data from the Land-Ocean temperature data output, so all 24 cases exhibited positive warming trends.

5 Discussion of Findings

The lines, `Month _ regression: gap size = ...`, indicate best fit lines for the frequency of records for each month. The 23 negative slopes in our output display an accelerated frequency of records, meaning that, while early record highs were separated by large gaps, recent records have occurred closer together. The lines `Month _ trend: deviation = ...`, indicate best fit lines for warming rates for each month. The 24 positive slopes indicate an overarching positive warming trend. There's a couple other interesting points:

- The Global Station data has higher warming rates (0.94-1.16) than the Land-Ocean data (0.75-0.896), and also has a more dramatic gap size acceleration. This might be because land heats faster than ocean, so the Land-Ocean data shows a weaker signal.
- December (Month 11) is the only month with a positive gap size slope. This might mean December records were frequent early on, or might stem from a unique data quirk. This detail is an outlier in an overall pattern of 23/24 accelerating records.

Metric	Land-Ocean	Global Station
Months with Decreasing Gap Sizes	11	12
Most Accelerating Gaps	June (-1.85)	August (-3.71)
Range of Warming Rates	0.75 – 0.89	0.94 – 1.16
Fastest Warming Month	February (0.89)	February (1.16)
Slowest Warming Month	June (0.75)	August (0.94)

Table 1: Key Values

To further visualize the test for global warming trends, shown in the Appendix: Warming Trends Over the Years for Each Month (section 7) below are the line plots for the deviations with respect to time. The x-axis displays 'Years' past 1880, so $x = 0$ indicates 1880 and $x = 145$ indicates 2025. The trend lines, fit to the monthly deviations, confirm that the climate not only changes but has a positive slope, indicating a long-term warming trend.

6 Conclusion

While Mr. Shah is correct that climate always changes, our statistical analysis distinguishes between two types of change: 1) random fluctuations that have always characterized the weather and 2) a direction towards warming we observe across all months. Our statistical analysis of 145 years of global temperature data provides compelling evidence for global warming. All months displayed a warming trend, and nearly all months displayed an acceleration of temperature records. The positive temperature trends across all twelve months in both datasets, and the negative gap slopes indicating accelerating record frequency in 23 of 24 cases, demonstrate that the climate system is undergoing a direction beyond natural fluctuations. The datasets have geographical diversification, and both displayed the same pattern, further supporting that climate is not stationary.

6.1 Limitations

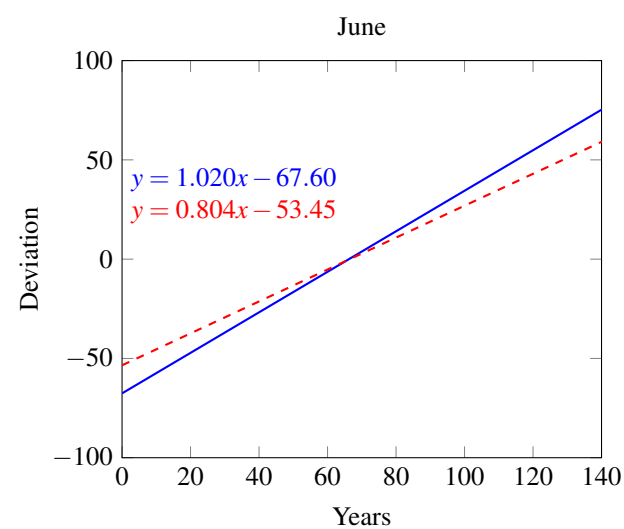
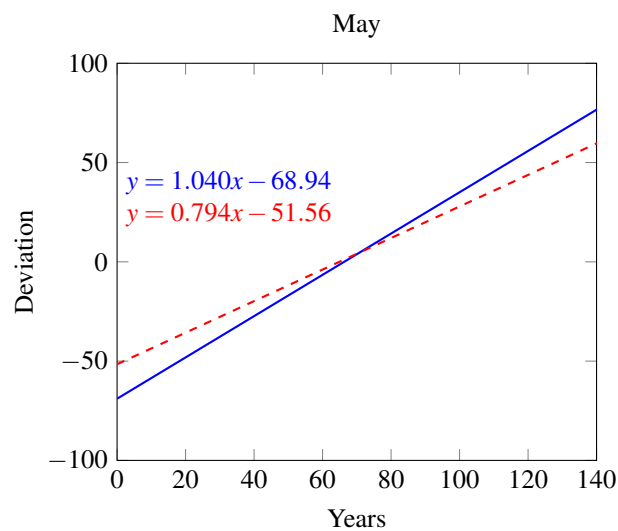
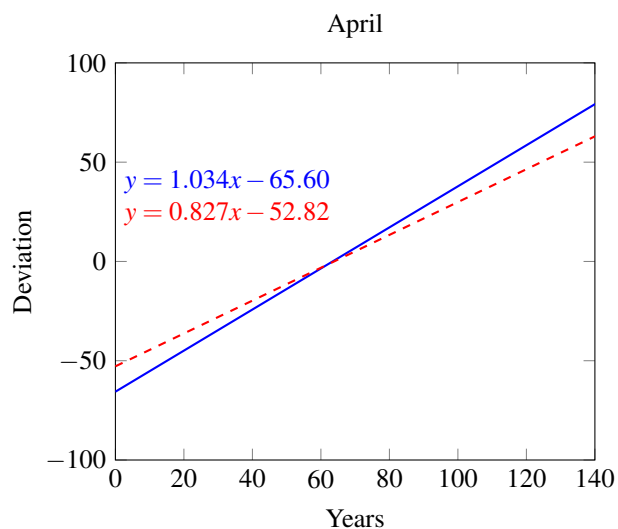
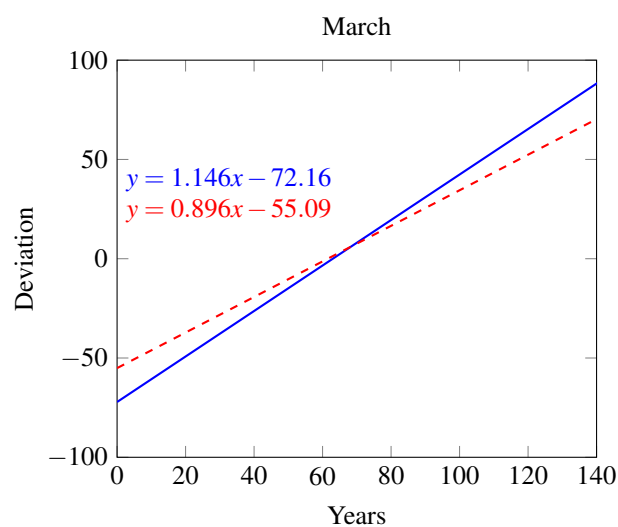
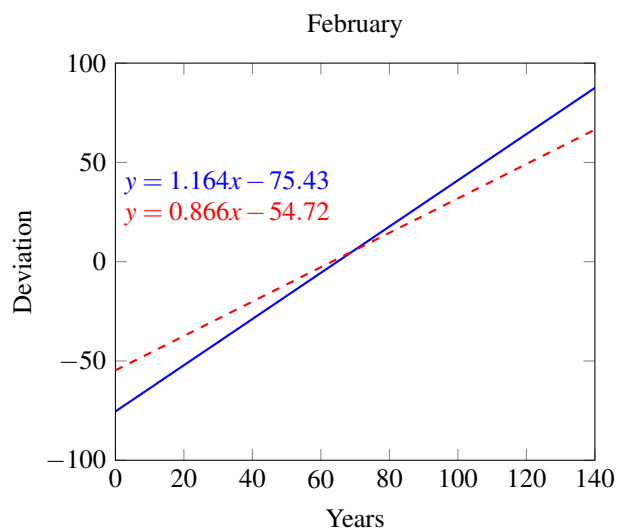
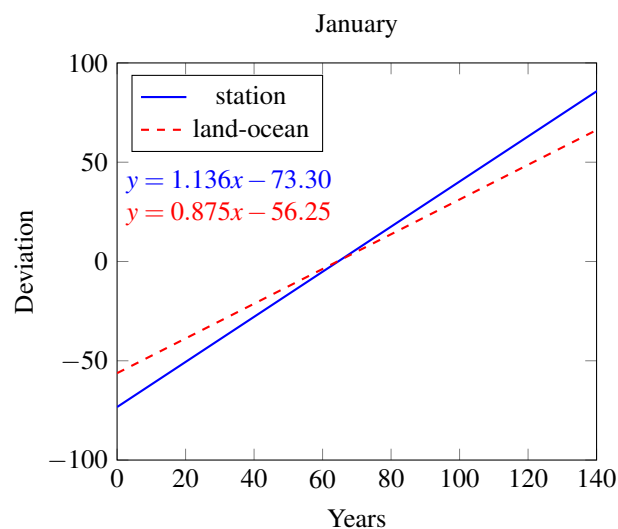
Despite our findings, our program has important limitations. For this project, we treated each month as an independent measurement, but in reality temperatures are not truly independent. Seasonal momentum carries and ocean heat persists for months. By treating months as independent, we cannot account for potential feedback loops, trends might appear more significant, and errors might be underestimated. Additionally, industrial activity was already significant by 1951, so our baseline (1951-1980 global average) might be warmer, skewed by fossil fuel emissions, potentially underestimating total temperature change. Our linear regression model worked well to detect a trend, but could not visualize an acceleration in temperature increase. We'd like to consider a nonlinear model and other visualization techniques (see below, 6.2).

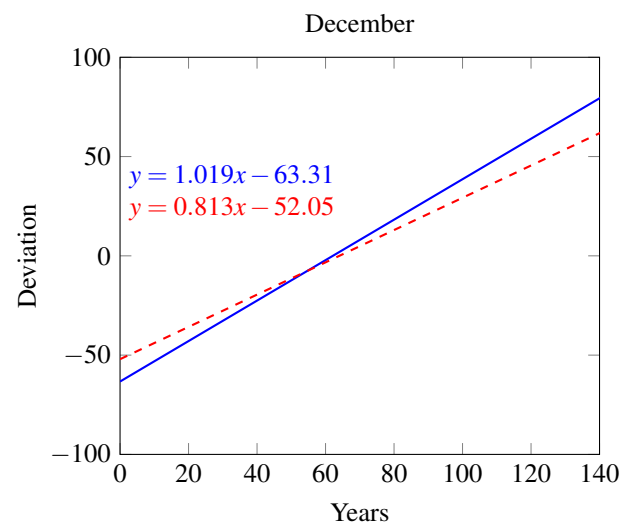
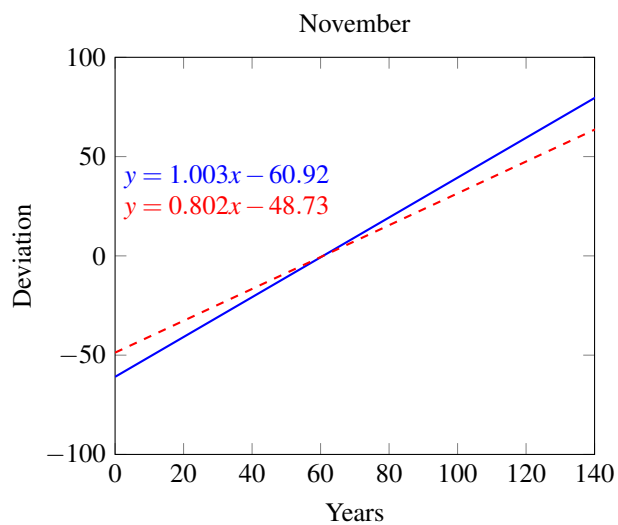
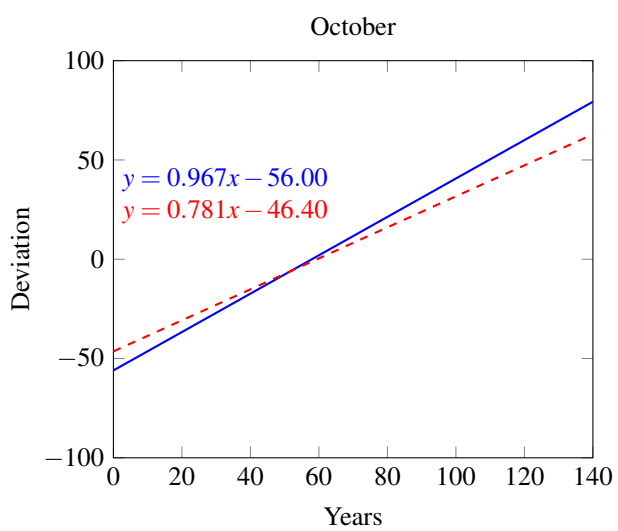
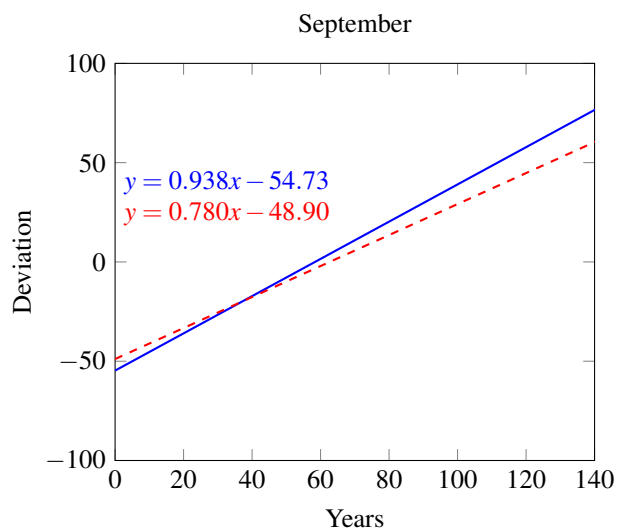
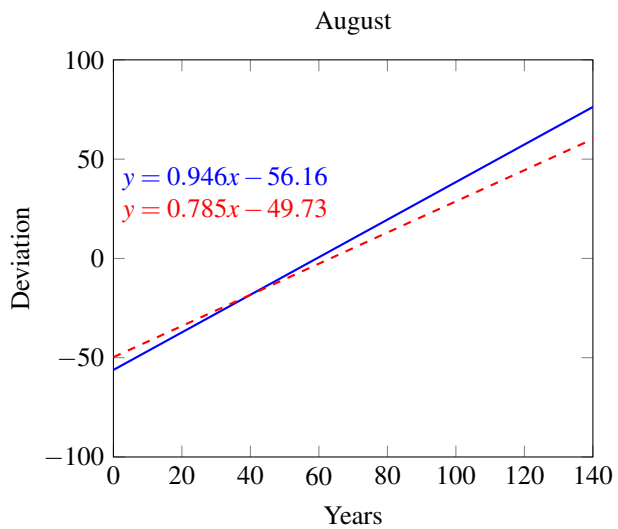
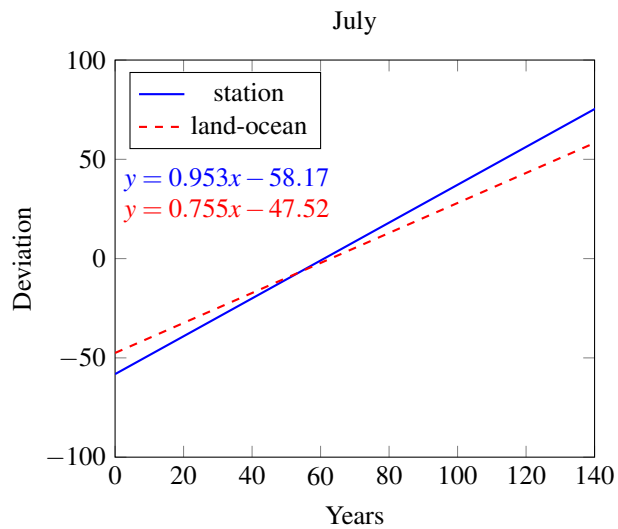
6.2 Potential Next Steps

We'd like to develop our linear regression model to better visualize climate trends. This could be achieved through a nonlinear model. Climate indicators like carbon dioxide concentration, ocean levels, and melting glaciers could better appear on our graphs. Incorporating this would make our model more realistic, as well as accurate, in depicting future climate patterns. We could also implement a piecewise function. Instead of creating one line for the 145-year period, an added piecewise could provide context by visualizing different slopes throughout time. This would help display the variations of climate change, as global warming does not occur at a constant rate. As mentioned in the limitations, historical events, like industrialization and heightened carbon emissions, contribute to the rate of global warming. By allowing for separate equations with separate slopes, we could capture when warming accelerates, how much it accelerates, and whether decades have different behaviors. This would not only help us more accurately predict global warming, but also highlight and scale the factors of our environment that contribute to it.

Ultimately, the climate data analysis goes beyond documenting trends of the past; it must inform future action. The increasing record-breaking and global warming we observe in this data set represent more than statistical patterns. They reflect fundamental changes to our planet that will continue to intensify without intervention. By refining our analytical methods through nonlinear modeling and piecewise functions, we can better identify inflection points where human intervention could potentially alter climate change. Understanding the rate and pattern of climate change is important to making policies for the future. Our analysis confirms: the climate is changing in unprecedented ways, and the window for effective action continues to narrow with each record-breaking month.

7 Appendix: Warming Trends Over the Years for Each Month





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9 Bibliography

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